

Within obstetrics specifically, PRP has previously been investigated in the context of caesarean section wound healing, where it was infiltrated intraoperatively at both the uterine incision (hysterotomy) margins and the subcutaneous skin layer; favourable effects were reported on skin-level scar quality in a single-centre randomized controlled pilot trial and a prospective observational study, although healing at the level of the myometrium itself was not separately assessed [13,14]. Consistent benefits have since been reported by other groups, further supporting a role for PRP in caesarean wound healing [15]. In parallel, we have also studied a different regenerative approach, topical lactic acid, specifically for episiotomy wound healing, through a randomized clinical trial of intravaginal lactic acid gel [16]. This dual line of work prompted us to examine the literature on PRP in episiotomy healing directly. The exercise proved instructive in an unexpected way: the answer returned by a conventional search of the major international databases differs materially from the answer returned once regionally indexed journals, trial registries and citation tracking are added. Any assessment of what is known about PRP in episiotomy repair therefore depends critically on how the literature is searched, and this observation motivates both the methodological approach and the framing of the present communication.

This asymmetry is not necessarily trivial to resolve by extrapolation. The caesarean incision and the episiotomy wound differ substantially in several respects relevant to healing biology: tissue composition and vascularity, degree and type of bacterial exposure (the episiotomy wound heals in immediate proximity to the vaginal and perianal microbiota), and the mechanical environment during the early postpartum period, where the perineum is subject to sitting, walking, and continuous activity that the caesarean incision is not [13–16]. These differences mean that favourable results obtained for PRP in caesarean wound healing cannot be assumed to generalize to the episiotomy wound without dedicated clinical evaluation.

It is also worth clarifying the scope of perineal trauma addressed here. Episiotomy is a deliberate surgical incision and constitutes only one form of perineal trauma; spontaneous intrapartum perineal lacerations (tears), which are not eliminated by a restrictive episiotomy policy, represent a related but distinct category. Episiotomy itself is not a single, uniform procedure: mediolateral and midline techniques carry different risks of extension toward the anal sphincter, and the broader intrapartum context, spontaneous versus operative (forceps- or vacuum-assisted) vaginal delivery, together with maternal, fetal, and constitutional factors, further shapes both the occurrence and the healing trajectory of perineal trauma more generally. The present communication, and the PRP-EpiHeal trial it introduces, focus specifically on the episiotomy wound rather than on spontaneous perineal lacerations or obstetric anal sphincter injury as separate entities; readers should bear this distinction in mind when considering the boundaries of the evidence described below.

In this communication we report a structured, reproducible search undertaken to determine whether and how PRP has been evaluated for episiotomy or intrapartum perineal wound healing; we appraise critically the small number of studies identified, including one randomized trial that is not retrievable through PubMed or Scopus; and we describe PRP-EpiHeal (ClinicalTrials.gov ID: NCT07669285), a two-arm randomized controlled trial designed by our group to provide prospectively registered, adequately powered and assessor-blinded evidence on this question.

2. Materials and Methods

A structured literature search was conducted to identify clinical studies evaluating platelet-rich plasma (PRP) for obstetric episiotomy or intrapartum perineal wound healing following vaginal delivery (spontaneous or operative), as distinct from gynecological perineal procedures. The search was designed to be reproducible; the full database-specific